# From Individual Predictions to Collective Forecasts: A Temporal-Graph Benchmark for Operational Cohorts

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## Abstract

Operational forecasts often concern groups of related entities, while individual event probabilities leave their collective uncertainty unspecified. We construct a temporal-graph forecasting benchmark for the joint first exits of the population active at a forecast origin, using a compute-cluster trace and a loan-application workflow. The benchmark separates individual-model selection from dependence modeling: tree and temporal neural predictors supply fixed marginal CDFs, while classical global, typed relation, and context-dependent factors generate coherent multihorizon entity outcomes. Controls include two dependence families, risk-stratified incidence permutation, and historical aggregate-error resampling. A frozen future-period evaluation covers 28 cluster origins and 61 workflow origins. The selected model reduces normalized count CRPS by 24.3% in the cluster and 8.3% in the workflow relative to independent sampling. Yet a declared reserve-loss proxy improves by 59.7% in the cluster and worsens by 9.6% in the workflow. Observed relation incidence contributes beyond individual risk, while richer context networks do not establish a consistent advantage. Simple aggregate residual resampling is the strongest workflow CRPS comparator. These findings support evaluating marginal bias, joint distributions, and decision loss separately. The contribution is a reproducible application benchmark and scalable reference implementation, with explicit limits on calibration, temporal evidence, and operational interpretation.

## 1 Introduction

A cluster operator sees a population of pending and running tasks. A workflow analyst sees a population of unresolved applications. In both cases, a useful question concerns their collective future: how many of the entities active now will remain in their current state, and how uncertain is that count? Related entities can evolve together because they belong to the same job, use the same machine, or share a recent handler. A collection of accurate individual probabilities does not specify this joint uncertainty.

Temporal graph learning provides useful representations of observed event histories. However, an individual prediction benchmark and an operational scenario benchmark answer different questions. Improving an entity's event probability need not improve the distribution of a group count. Conversely, adding shared random factors may improve a count distribution while leaving every individual probability unchanged. Even an improvement in a proper distributional score need not improve a particular asymmetric decision loss. An application study should measure these distinctions directly.

We construct a common forecasting task on two public operational logs: Google's 2011 cluster trace and BPI Challenge 2017. At a timestamp, an adapter reconstructs the currently observed entities and typed relation memberships. The target is their joint first-exit or resolution trajectory at four future horizons. The entire current population is evaluated, including entities that have not exited by the longest horizon. Future arrivals, eventual endpoints, and future assignments are not supplied to the forecaster. This yields a shared task across applications without assuming that a machine and a workflow handler represent the same physical constraint.

Our study separates individual forecasts from the coupling of their outcomes. We first compare tree predictors and temporal Mixer, GRU, and MLP ensembles. We then hold the selected individual CDFs fixed while comparing independent, global, typed relation, and context-dependent joint sampling. Classical Gaussian and common-shock factors make this comparison computationally feasible for changing populations. A risk-stratified incidence permutation tests the role of observed relations. A historical aggregate-error bootstrap provides a deliberately simple competitor that can correct group-level bias but does not generate a consistent entity-level joint law.

The contribution is an application benchmark and controlled empirical study, with reproducible adapters and a scalable reference implementation. It is not a new copula family, a discovery that operational entities are dependent, or a complete event-log simulator. The benchmark makes three questions separately testable: whether temporal graph information improves individual forecasts; whether observed relations improve collective forecasts when individual probabilities are identical; and whether any distributional improvement survives an operational loss and a future-period evaluation. Preserving negative answers is part of the benchmark's purpose.

## 2 Related work and scope

**Temporal prediction and generation.** GraphMixer demonstrates the strength of simple temporal encoders for link prediction [Cong et al., 2023]. We use an independently implemented Mixer-style survival adaptation, alongside recurrent and tabular controls, rather than claiming to reproduce its original link-prediction protocol. CEP3 forecasts community events and their times with a graph point process [Wang et al., 2022]; DG-Gen autoregressively generates continuous-time dynamic graphs [Hosseini et al., 2025]. These are relevant generative approaches, but their published tasks do not by themselves establish reliable distributions of current-cohort first exits. We do not report numerical superiority over these systems: a fair comparison would require explicit adaptations of outcome semantics, state initialization, horizon stopping, and candidate populations.

**Joint probabilistic forecasting.** TACTiS and TACTiS-2 learn copula-based multivariate predictive distributions and support irregular or unaligned time series [Drouin et al., 2022, Ashok et al., 2024]. Separating fitted marginals from a spatial Gaussian copula is also established, including in neural rainfall downscaling [Huk et al., 2025]. Our reference models use more restrictive dependence families, including the classical Marshall–Olkin shock construction [Marshall and Olkin, 1967]. The practical distinction is the use of current typed incidence to share random factors among changing entity populations, with identical individual forecasts across controls. We claim neither a universal modeling advantage over attentional copulas nor a new marginal–dependence decomposition.

**Operational simulation and uncertainty.** RIMS integrates learned predictions with business process simulation and runtime intercase information [Meneghello et al., 2025]. AgentSimulator models resource behavior and interactions [Kirchdorfer et al., 2025]. Closely related work reconstructs current process states and estimates state-conditional forecast-error intervals, including on BPI 2017 [Avramenko et al., 2026]. Thus, neither current-state initialization nor uncertainty estimation for operational forecasts is an unaddressed problem. Our study instead isolates joint outcome dependence under fixed individual CDFs, tests relation incidence, and compares multiple group projections and decision losses across a cluster and a workflow. The historical-error baseline reflects the relevance of direct aggregate uncertainty estimation; it is not a reproduction of the complete simulation system in Avramenko et al. [2026].

Resource-trace generation also has a distinct operational objective. Mimesys synthesizes executable workloads to reproduce resource contention [Kim et al., 2026]. Our sampled first-exit bins cannot substitute for executable workloads or full process simulations. They target the uncertainty of the observed cohort under the logged operating regime.

## 3 Joint forecasts for observed operational cohorts

At forecast origin $t$, let $V_t$ be the set of currently active entities and $\mathcal{H}_t$ the event history observable by $t$. Each entity has a current state $s_i(t)$ and zero or more observed relation memberships $g_r(i,t)$. Memberships may be shared across states. A missing membership creates no shared factor. Identities index the observed incidence structure; they are not learned identity embeddings.

For ordered horizons $h_1 < \cdots < h_K$, define $Y_{ik} = \mathbb{1}\{T_i \le t+h_k\}$, where $T_i$ is the first exit from the entity's current task state or, in the workflow application, its first qualifying application-resolution event. We estimate the conditional joint law of these outcomes for the entire current cohort. This is a discrete-horizon forecast: each draw assigns an exit bin or survival beyond $h_K$. It does not specify within-bin event times, subsequent returns, or outcomes for future arrivals.

An individual forecaster supplies a monotone CDF $F_{ik} = \Pr(T_i \le t + h_k \mid \mathcal{H}_t)$. Individual predictive accuracy does not determine collective uncertainty. For a known group $A \subseteq V_t$, its exit count is $C_{Ak} = \sum_{i\in A} Y_{ik}$ and

$$\begin{aligned}\operatorname{Var}(C_{Ak} \mid \mathcal{H}_t) = &\sum_{i\in A} F_{ik}(1-F_{ik}) \\ &+ 2 \sum_{i<j,\, i,j\in A} \operatorname{Cov}(Y_{ik}, Y_{jk} \mid \mathcal{H}_t).\end{aligned} \tag{1}$$

This standard decomposition motivates a controlled comparison: keep every $F_{ik}$ fixed and vary only the joint sampling law.

### 3.1 Observed-incidence factor sampling

For each forecast draw, independently sample a global factor $Z_0$, one factor $Z_{r,g}$ per observed relation group, and a private factor $\epsilon_i$ per entity. The Gaussian construction is

$$\begin{aligned} Z_i &= a_i Z_0 + \sum_{r=1}^{R} b_{ir} Z_{r,g_r(i,t)} + c_i \epsilon_i, \\ a_i^2 + \sum_r b_{ir}^2 + c_i^2 &= 1, \qquad U_i = \Phi(Z_i). \end{aligned} \tag{2}$$

All underlying factors are standard normal. The nonnegative loading parameterization models positive residual association. It cannot express general negative dependence. When a relation is unavailable, its squared loading is transferred to the private component. Consequently, each $U_i$ is uniform, independently of the cohort size and incidence pattern. Setting $Y_{ik}^{(m)} = \mathbb{1}\{U_i^{(m)} \le$

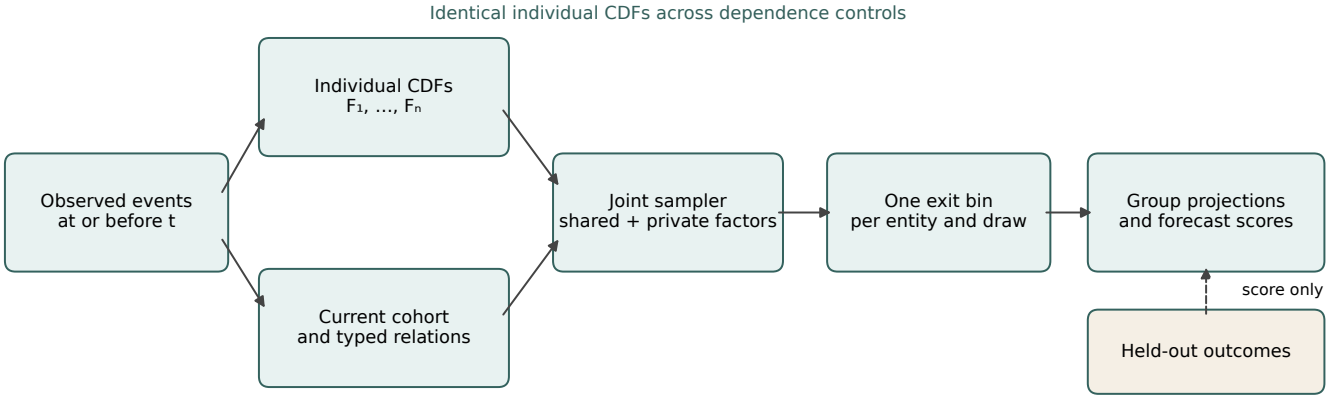



Figure 1: Inference contract at a forecast origin (schematic). Observed history supplies the current cohort, relations, and individual CDFs. Shared randomness changes the joint law while preserving those CDFs. Future outcomes enter scoring only. Training uses separate earlier chronological windows.

$F_{ik}\}$ preserves the supplied marginals and makes every sampled trajectory monotone across horizons. The same shared draws are reused across streamed entity chunks.

We also evaluate a classical common-shock construction related to Marshall–Olkin models [Marshall and Olkin, 1967]. With independent unit-rate exponential factors $E_0, E_{r,g}, E_i$ and nonnegative weights summing to one,

$$L_i = \min\left\{\frac{E_0}{a_i^2}, \left\{\frac{E_{r,g_r(i,t)}}{b_{ir}^2}\right\}_{r=1}^{R}, \frac{E_i}{c_i^2}\right\}, \qquad (3)$$
$$U_i = 1 - \exp(-L_i).$$

A zero-weight factor has an infinite race time. Since the minimum has unit exponential rate, this construction also preserves each individual CDF. Neither construction is a new copula family or an identified causal shock model.

The independent comparison sets the private weight to one. The global model learns state-specific global/private weights. The static relation model adds state-specific weights for each relation type. The context model adds a two-layer, width-32 MLP offset to the state-specific logits, with bounded offsets and a quadratic shrinkage penalty. A softmax allocates variance weights, with at least 0.02 private mass. This prevents perfect dependence in fitted models. Sampling costs are linear in the number of entities and observed memberships per draw; no dense entity-by-entity covariance matrix is formed.

### 3.2 Fitting without in-sample forecast residuals

We fit dependence on chronological forward-fold predictions: every training CDF was produced by an individual model trained before that target period, with any calibrator fitted in a separate earlier window. At each origin we sample uniform entity pairs, uniform eligible relation-group pairs, and group-size-weighted pairs. Sampling uses observed membership and group size, not future outcomes. Pairs may span current states.

Training minimizes the composite negative log probability of the observed binary pair, with independently sampled horizon indices for its two entities. The Gaussian pair law uses the correlation implied by shared factors and a bivariate normal CDF; the common-shock pair law has an analytic expression. Validation uses a fixed balanced pattern over the $K^2$ horizon pairs. Pair likelihood is a tractable fitting objective; it does not guarantee optimal aggregate scores or operational decisions. Dependence parameters can absorb individual-model misspecification as well as residual association.

### 3.3 A control on relation incidence

For a fitted static model, we permute relation memberships within current state and half-unit logit-risk bins of the last-horizon marginal prediction. Known group sizes within each stratum, all individual CDFs, and fitted loading weights are preserved. Evaluation groups retain their true observed memberships. Thus, only the identity of entities sharing relation noise changes. A benefit from true incidence supports the predictive usefulness of this grouping under the fitted model; it does not establish that changing an actual job or handler assignment would cause the same effect.

## 4 Evaluation quantities

For each origin, we score current-state population counts and observed relation groups containing at least five entities. Scores are normalized by the group's known current size. We average equally across groups within a projection, horizons, projection types, and forecast origins. We

also report projections separately so a population-level benefit cannot conceal a relation-level loss.

For $M$ independent scenario draws, the fair ensemble CRPS [Ferro, 2014, Gneiting and Raftery, 2007] is

$$\widehat{\mathrm{CRPS}}(X, y) = \frac{1}{M}\sum_{m} |X^{(m)} - y| - \frac{\sum_{m \neq n} |X^{(m)} - X^{(n)}|}{2M(M-1)}. \tag{4}$$

We additionally report 90% central interval coverage, width, and interval score; an energy score for each group's normalized multihorizon trajectory; and CRPS for the fraction of groups in which at least 20% of the current cohort exits. These are evaluations of specified operational projections, not a claim that a small collection of scores identifies the full high-dimensional law.

To expose decision tradeoffs, let $R_{Ak} = |A| - C_{Ak}$ be the number of current members that have not exited. For an under/over penalty ratio $r$, select the $r/(r+1)$ forecast quantile $b$ of $R_{Ak}$ and evaluate

$$\ell_r(b, R_{Ak}) = \frac{r(R_{Ak} - b)_+ + (b - R_{Ak})_+}{|A|}, \quad r \in \{3, 9\}, \tag{5}$$

where $(x)_+ = \max(x, 0)$. This is a reserve-cost proxy for the current cohort. It is not measured currency, a staffing policy, or a counterfactual deployment estimate. Future arrivals and subsequent state returns are outside its scope. Distributional score improvements and improvements under this particular loss are reported separately.

We average random-seed repetitions within each forecast origin before paired time-block resampling. Seeds and individual tasks are not additional independent temporal observations. The final analysis uses circular moving blocks of one and two days for the cluster trace and 14 and 28 days for the workflow trace, reporting when the available span is too short to support a chosen block size. The resulting intervals are descriptive approximations for finite dependent traces, rather than guarantees of performance at other sites.

# 5 Datasets and experimental design

## 5.1 Two operational adapters

**Compute cluster.** We use all 500 task-event shards of Google ClusterData2011-2 [Google, 2011]. The trace covers approximately 29 days in one cluster. Positive trace timestamps use the publisher's 600-second physical origin; time-zero rows describe an initial state with unknown preceding age. The adapter replays observed events in timestamp and source order. Tasks are inactive, pending, or running. The prediction target is the first change of the current pending/running state, at 1, 5, 15, and 60 minutes. It is not necessarily task completion: a running task may fail, be evicted, or be killed. Current job membership and known running-machine membership define two relation types. A pending or unknown-machine task has no shared machine factor. Missing requests retain known preceding values, with missingness indicators; publisher imputation and unknown initial age are retained as cutoff features. Requests are not measurements of resource utilization.

**Loan application workflow.** BPI Challenge 2017 records applications initiated during 2016 with later follow-up [van Dongen, 2017]. We predict the first occurrence of `A_Pending`, `A_Denied`, or `A_Cancelled` after creation, at 1, 3, 7, and 14 days. "Resolution" denotes this declared milestone; it does not mean repayment or necessarily the last logged event. Cases with a qualifying prior resolution are no longer members of the active cohort. The relation is the last observed workflow handler, updated from supported workflow lifecycle events. This is historical association, not a claim that all associated cases exclusively occupy the handler's capacity. Retrospective trace attributes, eventual offer outcomes, and amounts without an availability audit are excluded. Activities and lifecycle categories unavailable in the training vocabulary map to an unknown token.

## 5.2 Chronological partitions

All features are built from events at or before the forecast origin. Labels are attached afterward. Entity identifiers are storage keys, not learned embeddings. Ongoing entities carry their observed prefix across partitions; their future outcomes are never used as prefix features. The longest horizon must lie within the relevant label window.

For Borg, individual-model training ends at elapsed day 18, calibration uses days 18–20, and development targets use days 20–22. Dependence fitting uses forward predictions in days 8–10, 12–14, and 16–18; the corresponding individual models train before days 6, 10, and 14, with the following two days reserved for calibration. Final forecast origins are every six hours from day 22 through day 28.75: 28 complete current cohorts.

For BPI, individual-model training ends before September 2016, calibration uses September, and development targets use October. Forward predictions for dependence fitting use June, July, and August, each with an earlier training and calibration period. Final forecast origins are noon UTC each day from November 1 through December 31: 61 complete current cohorts. Follow-up through January 15 supports all 14-day labels. The archive concerns 2016 applications; this experiment does not require predicting unobserved 2017 arrivals.

The final evaluation contains 3,760,001 task–origin queries in Borg and 114,051 case–origin queries in BPI.

Borg cohorts range from 105,077 to 171,603 active tasks; BPI cohorts range from 1,484 to 2,189 active cases. These counts include repeated entities at different origins and are not independent sample sizes. The final Borg prefix contains 144,648,283 task-event rows.

Model selection was exploratory and development-informed. Before opening final labels, we froze model choices, checkpoints, code hashes, origins, metrics, random seeds, and uncertainty calculations in an immutable manifest. No model is retrained or recalibrated on final outcomes. This is a development-informed holdout freeze, not prospective preregistration of the initial research idea.

### 5.3 Individual models and common-CDF controls

Histogram gradient boosting (HGB) uses personal, global, and relational feature arms. The personal arm describes observed state, age, event history, and available requests or offer history. The global arm adds calendar features and system-wide activity; the relational arm additionally summarizes current job/machine populations or last-handler case history. These arms are nested. Borg has 32 base features and an extension with 144 recent-event counts; BPI has 103 features. Four monotone calibration variants are compared: sorted raw horizon probabilities, independently Platt-calibrated and sorted probabilities, and two shared positive-slope Platt maps with population or equal-origin fitting weights. Temporal encoders receive the same extended tabular features and the last 16 observed events per scope: task, job, and machine for Borg; case and handler for BPI. Entity identifiers are excluded from learned embeddings. The temporal Mixer and GRU use a five-category first-exit head; the MLP uses the tabular features alone. We compare raw and temperature-calibrated three-seed mean predictions. Selection minimizes equal-origin integrated Brier score, separately by current state, without choosing the best seed. Here integrated Brier means the arithmetic mean of binary Brier scores at the four specified horizons, with equal forecast-origin weights; it is a discrete-horizon summary rather than a continuous-time integral.

All dependence controls share this selected CDF. Each family is fitted with seeds 0, 1, and 2, and evaluated with 512 draws per origin. Seed 0 is also evaluated with 1,024 draws as a Monte Carlo sensitivity. Both families and all controls remain in the final tables even when development selection favors a single configuration. Hyperparameters and complete development comparisons are reported in the appendix and machine-readable manifests.

### 5.4 Historical aggregate-error comparator

For each development origin and eligible group, let $\mu_{Ak} = \sum_{i \in A} F_{ik}$ and $s_{Ak} = \sqrt{1 + \sum_{i \in A} F_{ik}(1 - F_{ik})}$. We store the four-dimensional residual $e_A = (C_A - \mu_A)/s_A$. A draw selects one historical origin for all groups, then a residual vector from a group with the closest $\lfloor \log_2 |A| \rfloor$ size bin within the same projection. Population states retain their own residual vectors. Predicted counts $\mu_A + s_A e_A$ are clipped to $[0, |A|]$, rounded, and made nondecreasing across horizons. Development evaluation leaves out the scored origin; final evaluation uses all development residuals with no further fitting. This comparator can change group means and uses development labels as fitting data. Its separate count projections need not be realizable as a single joint entity scenario. Accordingly, it is an aggregate-forecast benchmark, not a matched-marginal dependence ablation.

## 6 Results

### 6.1 Individual-model selection does not select a joint distribution

Table 1 reports the development comparison used to select individual forecasts. Borg selects the original HGB global arm for pending tasks (integrated Brier 0.06415) and the relational arm for running tasks (0.01591). The additional recent-event-count features and the neural alternatives do not displace these choices. BPI selects the temperature-scaled three-seed temporal Mixer ensemble (0.09796), ahead of the best HGB candidate (0.10039). These are development selection results, not an unbiased ranking on final data.

We now hold each selected CDF fixed across dependence controls. Table 2 reports all frozen final models; Figure 2 shows the distinction between count CRPS and reserve loss. The two datasets are not pooled into a single score. They have different event semantics, horizons, group structures, and observation spans.

### 6.2 Cluster: shared variation matters, with an additional relation signal

Across 28 final cluster origins, the development-selected Gaussian static model reduces macro count-fraction CRPS from 0.03417 to 0.02587, a 24.3% improvement over independent sampling. The mean paired difference is $-0.00829$, with a one-day block interval $[-0.01056, -0.00611]$ and a two-day interval $[-0.00985, -0.00686]$. At reserve penalty ratio 9, loss falls from 0.13106 to 0.05280, a 59.7% reduction in this declared proxy. Its paired one-day interval is $[-0.11744, -0.04280]$ for a mean difference of $-0.07826$. These are within-trace estimates over seven final days,

Table 1: Development integrated Brier used for model selection. Each row selects its calibration variant on development; neural rows average three seeds before scoring. These are selection scores, not final estimates. HGB history is the additional recent-event-count feature arm.

| Individual predictor | Borg pending | Borg running | BPI |
|---|---|---|---|
| HGB personal | 0.07574 | 0.01599 | 0.10039 |
| HGB global | 0.06415 | 0.01632 | 0.10314 |
| HGB relational | 0.06586 | 0.01591 | 0.10251 |
| HGB history personal | 0.07613 | 0.01600 | – |
| HGB history global | 0.06900 | 0.01653 | – |
| HGB history relational | 0.06599 | 0.01609 | – |
| MIXER ensemble | 0.06928 | 0.01670 | 0.09796 |
| GRU ensemble | 0.06690 | 0.01676 | 0.09871 |
| MLP ensemble | 0.06789 | 0.01792 | 0.10320 |

Table 2: Frozen final evaluation, averaged over forecast origins. Fitted dependence models use three seeds; the historical-residual sampler uses one fixed seed. CRPS and reserve loss are normalized by group size (lower is better); coverage targets 0.9. G: Gaussian; S: common shock. Independent draws are shown for each family to expose Monte Carlo variation. The historical-residual comparator changes aggregate means and lacks entity-level coherence.

| Model | Borg CRPS | Cov. | $\text{Loss}_9$ | BPI CRPS | Cov. | $\text{Loss}_9$ |
|---|---|---|---|---|---|---|
| G: Independent | 0.03417 | 0.659 | 0.1311 | 0.05010 | 0.536 | 0.1177 |
| G: Global | 0.02646 | 0.980 | 0.0551 | 0.04595 | 0.890 | 0.1290 |
| G: Typed static | 0.02587 | 0.906 | 0.0528 | 0.04649 | 0.793 | 0.1222 |
| G: Typed context | 0.02563 | 0.915 | 0.0541 | 0.04630 | 0.828 | 0.1258 |
| G: Permuted incidence | 0.02691 | 0.892 | 0.0586 | 0.04694 | 0.776 | 0.1194 |
| S: Independent | 0.03416 | 0.660 | 0.1310 | 0.05013 | 0.535 | 0.1178 |
| S: Global | 0.02769 | 0.848 | 0.0666 | 0.05095 | 0.579 | 0.1176 |
| S: Typed static | 0.02646 | 0.861 | 0.0639 | 0.05026 | 0.607 | 0.1178 |
| S: Typed context | 0.02611 | 0.866 | 0.0629 | 0.05028 | 0.680 | 0.1179 |
| S: Permuted incidence | 0.02719 | 0.852 | 0.0670 | 0.05003 | 0.593 | 0.1172 |
| Historical residual | 0.03089 | 0.875 | 0.0840 | 0.04387 | 0.831 | 0.1258 |

not independent-task confidence bounds or measured deployment savings.

Most of the CRPS gain already appears with a global factor: its score is 0.02646. Adding typed relation factors improves that by a further 2.2%, with paired difference $-0.000583$ and one-day interval $[-0.001043, -0.000185]$. At ratio 9, the corresponding global-to-static loss change is 0.05506 to 0.05280. The large independent-to-static improvement therefore must not be attributed entirely to graph relations.

The incidence control localizes the additional relation signal. Permuting memberships increases static-model CRPS from 0.02587 to 0.02691. The paired difference for true minus permuted incidence is $-0.001031$, with one-day interval $[-0.001175, -0.000873]$. Job-level CRPS changes most: 0.01451 with true incidence versus 0.01729 with permuted incidence. Machine-level CRPS changes from 0.02504 to 0.02535; population CRPS is essentially unchanged (0.03807 versus 0.03808). Under this controlled sampler intervention, correct job grouping carries predictive information beyond individual event risk and global random variation.

Macro coverage requires a qualification. Static Gaussian coverage averages 0.906 across projections, compared with 0.659 for independence. However, static population coverage is only 0.763, while job and machine coverage are 0.978 and 0.976. Its near-nominal macro value therefore does not establish calibration within every projection. Independence is particularly unreliable for the population count, with coverage 0.100. This projection breakdown is more informative than a single aggregate coverage number.

The context model obtains a slightly lower mean CRPS, 0.02563, but its context-minus-static difference has one-day interval $[-0.000661, 0.000195]$. Its reserve loss is also slightly higher, 0.05413. The evidence does not establish a consistent benefit from the additional context network. Common-shock static factors improve over their independent control (0.02646 versus 0.03416), but are weaker than Gaussian static factors on the principal mean scores. The historical-residual comparator has CRPS 0.03089 and reserve loss 0.08400. It is useful as an aggregate-error baseline, but it does not explain away the selected model's cluster improvement.

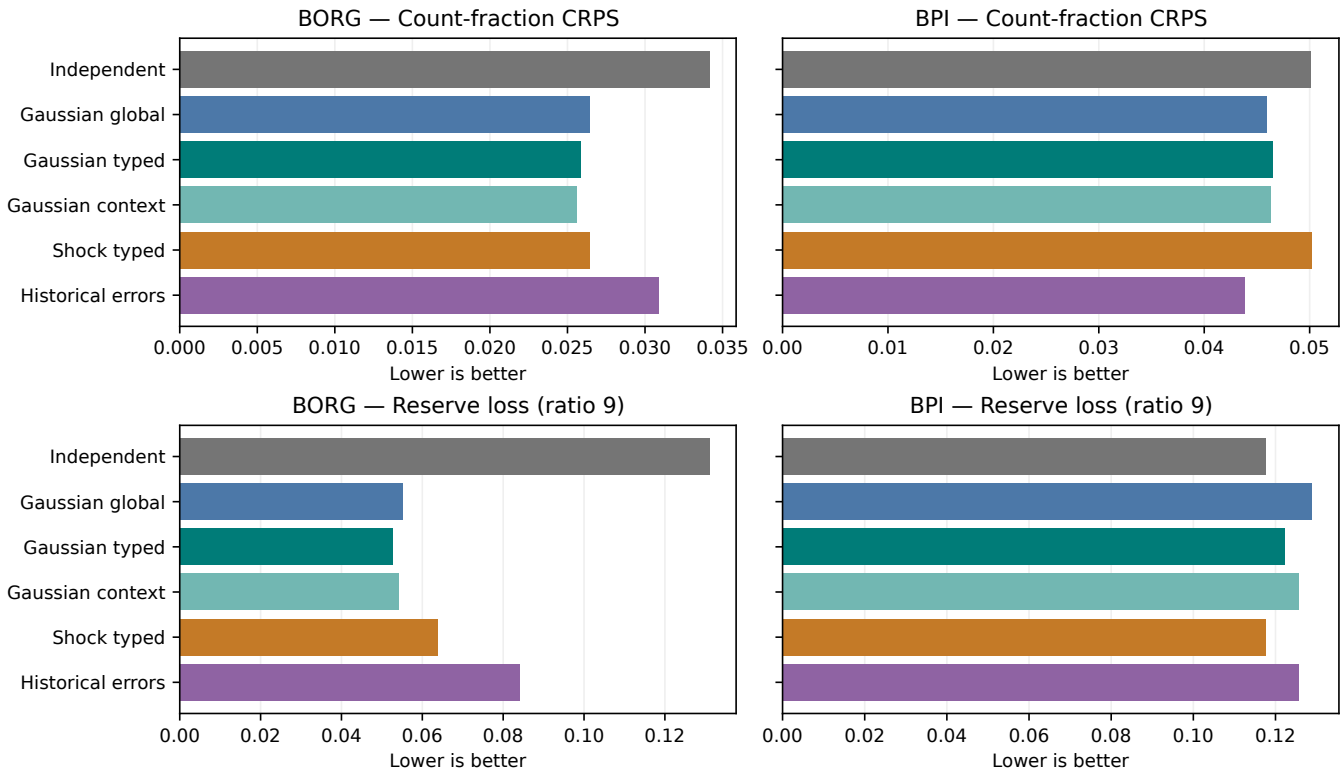



Figure 2: Final count-fraction CRPS and reserve loss for representative controls. The same individual CDFs underlie the Gaussian and common-shock comparisons. Historical errors are a separate direct aggregate forecast. Lower values are better on each axis; the two axes measure different objectives.

### 6.3 Workflow: distributional improvement and decision loss diverge

On the 61 frozen final workflow origins, the development-selected Gaussian global model reduces macro count-fraction CRPS from 0.05010 to 0.04595, an 8.3% improvement over independent sampling. Its mean paired difference is $-0.00415$; the 14-day block interval is $[-0.00638, -0.00188]$, and the 28-day interval also excludes zero. Mean 90% coverage rises from 0.536 to 0.890, while normalized interval width rises from 0.140 to 0.222. This illustrates a practical failure of independent sampling: reasonable individual probabilities can induce overly concentrated collective forecasts.

The projection breakdown localizes the effect. Population CRPS improves from 0.03421 to 0.02728 and population coverage from 0.219 to 0.869. Handler-group CRPS improves more modestly, from 0.06599 to 0.06462, with coverage rising from 0.853 to 0.911. Equal projection weighting keeps these effects visible rather than allowing the many handler groups to obscure the population result.

That improvement does not extend to the specified reserve loss. At under/over ratio 9, the selected model's loss is 0.12899 versus 0.11770 for independence, a 9.6% increase. The paired difference is 0.01129; its 14-day block interval is $[-0.00117, 0.02129]$, while the 28-day interval is $[0.00426, 0.01746]$. The direction of the mean tradeoff is clear, but its uncertainty depends on the block choice. We therefore report it as a held-out decision-loss tradeoff, not a universally established cost penalty. At ratio 3, the mean loss likewise increases, from 0.08933 to 0.09873.

Observed handler incidence provides a smaller, distinct signal. Gaussian static factors obtain CRPS 0.04649, compared with 0.04694 after risk-stratified incidence permutation. The difference, $-0.000454$, has a 14-day block interval $[-0.000726, -0.000188]$. However, the corresponding reserve loss increases from 0.11936 under permutation to 0.12219 under true incidence; the paired interval for this increase is $[0.00119, 0.00442]$. Thus, even the same relation-incidence control can favor one model under a distributional score and the other under a decision loss. It does not identify a causal effect of actual handler assignment.

The richer context model does not establish a reliable improvement over the static model: its mean CRPS difference is $-0.000188$, with a 14-day interval $[-0.000626, 0.000262]$. The common-shock static model has CRPS 0.05026 versus 0.05013 for its independent control. Merely adding a dependence model is therefore insufficient, and the choice of family matters.

The direct historical-residual comparator achieves the lowest aggregate CRPS among these workflow compar-

isons, 0.04387, while its coverage is 0.831 and its ratio-9 loss is 0.12576. It can shift aggregate means and is not a matched-marginal ablation. Its mean CRPS advantage over the selected Gaussian model is not resolved by the block intervals: the Gaussian-minus-residual difference is 0.00208, with a 14-day interval $[-0.00070, 0.00536]$. This distinction is consequential: at 14 days, the selected individual ensemble predicts an average population resolution fraction of 0.5252 versus the observed 0.5792. Every matched-marginal copula inherits that mean bias. The residual baseline's advantage is evidence for retaining simple aggregate-error controls, rather than evidence that its separately sampled projections form a better entity-level joint law.

### 6.4 Marginal diagnostics and data-quality strata

Figure 3 shows that dependence comparisons inherit distinct marginal errors. At 60 minutes, mean Borg pending-task exit probability is 0.3193 versus an observed fraction of 0.3184. For running tasks the corresponding values are 0.1199 and 0.1287. Their final integrated Brier scores are 0.08251 and 0.02937. The BPI 14-day bias is larger in absolute fraction, as discussed above. A small pooled mean error is not evidence of conditional calibration at every origin or relation group; the released reliability and origin tables retain this variation.

Cutoff-defined strata also matter. About 21,876 running tasks per final origin have unknown initial state age. Their integrated Brier is 0.00025, compared with 0.03529 for known-age running tasks. Long-lived initial-state entities are thus a relatively easy subpopulation, and pooled scores conceal that heterogeneity. The unknown-age pending stratum contains only one task per origin and cannot support a general robustness claim. Running tasks with a known preceding publisher-imputed event have Brier 0.01078 versus 0.03006 without one; these are descriptive case-mix differences, not causal effects of imputation. All strata remain in the primary complete-cohort evaluation.

### 6.5 Numerical sensitivity and execution

Doubling seed-0 scenario draws from 512 to 1,024 changes macro CRPS by at most 0.000122 across the two applications, families, and five controls. Within each application and family, the ordering of the five controls is unchanged. This checks the reported CRPS comparisons against finite-ensemble variation; it does not replace uncertainty over future forecast origins.

On the shared RTX A6000 host, each 512-draw Borg evaluation of all five controls across 28 origins takes 245.8–251.0 seconds wall time, including sampling and scoring. The corresponding BPI evaluation across 61 origins takes 3.48–3.68 seconds. Borg peak allocated PyTorch memory is at most 0.67 GiB per evaluation process and peak host resident memory is 2.05 GiB. These are evaluation costs with individual CDFs already computed, measured while other evaluations share the device; they exclude data reconstruction, training, and individual-model inference. They are not an isolated latency benchmark.

The final artifact audit verifies all 264 frozen file hashes, every final CDF and label trajectory, the exact frozen feature transform, and the expected origins and comparisons. No frozen model or evaluation file was modified during final evaluation. The separately audited lossless timestamp correction is described in the appendix.

## 7 Discussion and limitations

**What a collective forecast adds.** The factor sampler provides a distribution over entity outcomes that can be projected onto several overlapping groups without refitting a separate count model. This is useful when the same task belongs to a job and a machine, or when an analyst changes the grouping after generation. The historical-residual comparator offers a different advantage: it can adjust aggregate bias directly. Its projections do not, in general, arise from a common set of entity outcomes. Neither property alone guarantees better predictive scores. The experiments compare these practical tradeoffs under a declared target and operating regime.

**Marginal bias is not fixed by a copula.** Every matched-marginal dependence control has the same expected group count, $\mathbb{E}[C_{Ak}] = \sum_{i \in A} F_{ik}$. Changing shared factors can change dispersion and tails, but cannot repair a systematically biased mean. Apparent improvements in a distributional score may therefore coexist with a worse decision quantile. Moreover, composite-likelihood fitting can attribute common forecast errors to dependence. A learned global or relation factor is a predictive construction, not proof of a physical common cause.

The score–loss disagreement is not a theoretical paradox. CRPS aggregates quantile losses across probability levels [Gneiting and Raftery, 2007], whereas a particular reserve penalty evaluates one quantile. Improving the former does not guarantee improvement at every level. The application evidence makes this distinction operational: validation should report the intended loss alongside distributional scores, and inspect mean bias before interpreting a wider scenario distribution as a useful decision forecast.

**Scope of the evidence.** The two logs are single organizational settings, and the final cluster period contains only seven days. Millions of entity queries do not create millions of independent temporal replications. The block intervals describe variation within these finite traces;

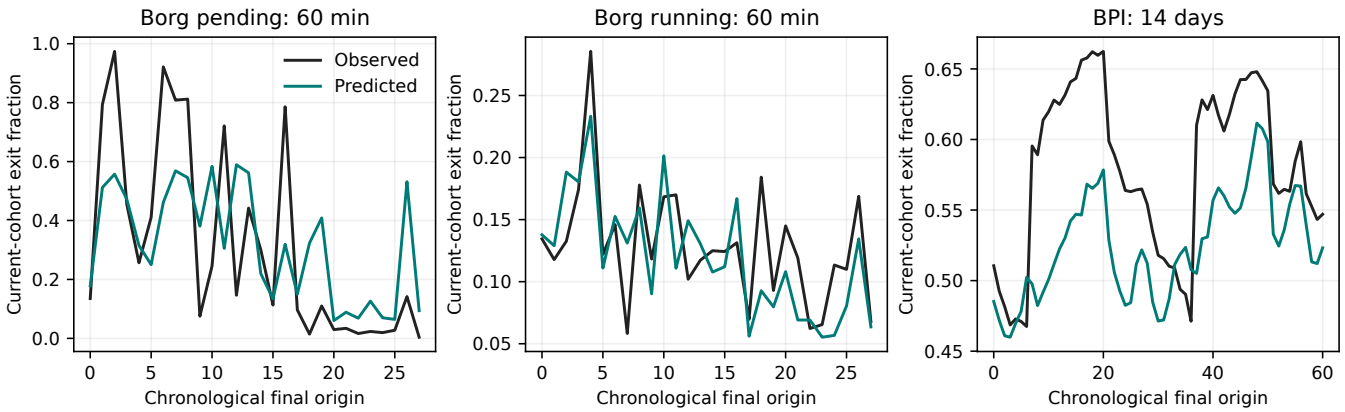



Figure 3: Longest-horizon population mean forecasts and outcomes at each final origin. These means are identical in expectation for all matched-marginal dependence models. A copula changes dispersion and tails, but cannot repair their mean forecast bias.

they cannot establish universal superiority or reliability in a new cluster or bank. The relatively short development periods also constrain tuning and historical residual estimation. Architecture, width, history length, and pair sampling are practical choices rather than exhaustive hyperparameter optima. Published point-process and full simulation systems remain important unimplemented numerical comparisons.

The factor families model positive association and use memberships observed at the origin. They do not explicitly simulate future membership changes, arrivals, repeated transitions, resource allocation, or counterfactual policies. Dependence across horizons is restricted by one latent event time per entity. First exits can be operationally useful while still being an incomplete description of total future workload. BPI last-handler groups include administrative/system behavior and do not identify exclusive worker occupancy. The normalized reserve loss is a transparent forecast-based proxy, not measured economic savings or a staffing recommendation.

Equal weighting of projection types and size-normalized groups is a benchmark convention. It prevents large populations from automatically dominating the comparison, but it is not an estimated organizational utility. Different group weights, penalties, or target horizons can change the ranking. The projection and horizon tables are therefore necessary context for the macro results.

**Implications for scientific dataset generation.** The reusable artifact is a timestamp-bounded adapter, an explicit outcome contract, and a collection of scenario diagnostics. A future dataset-generation system should preserve these contracts while adding arrivals and subsequent state transitions. Generating a visually plausible event stream is insufficient evidence of faithful operational uncertainty. Such a system would also need downstream utility, privacy, and policy-validity experiments; none are claimed by this study. The present data are public research releases, and original license terms and attribution remain attached to their sources.

# 8 Conclusion

Current-cohort forecasting creates a practical bridge between temporal event prediction and operational uncertainty. Across a cluster and a workflow, holding individual probabilities fixed reveals substantial benefits from shared variation, a smaller but measurable contribution from observed relations, and no consistent advantage from richer context networks. These benefits depend on the application and the evaluation target: better count CRPS can coexist with worse reserve loss, and simple aggregate-error resampling remains competitive. The benchmark and reference implementation make these distinctions reproducible. Extending the task to arrivals, repeated transitions, and additional organizations is needed before treating such scenarios as a general operational simulator.

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# A Implementation and reproducibility

**Sampling individual training queries.** Borg individual-model queries are built hourly, sampling up to 2,000 active tasks without replacement from each state, with inverse inclusion weights. Sampling uses a fixed seed and never uses future labels. Complete-population scenario evaluation uses every active task at six-hour origins. BPI queries contain every unresolved case at daily origins. Consequently, the individual development table and the complete-population scenario table are different estimands; the latter is not obtained by scoring a small sampled population. The bounded development adapters produce 1,833,102 Borg queries and 519,142 BPI queries across training and development periods. The pre-final Borg prefix contains 115,400,646 task-event rows; the BPI prefix contains 976,753 events.

**Marginal training.** HGB uses 250 boosting iterations, at most 31 leaves, a minimum of 100 samples per leaf, learning rate 0.05, and L2 regularization 1.0, with early stopping disabled. The recent Borg feature extension counts nine event types over 1/5/15/60-minute windows for personal, global, job, and machine scopes, using log-one-plus counts. Temporal encoders use width 64, two Mixer blocks or a GRU, typed event categories, continuous event features, observed time lags, and padding masks. Histories are ordered oldest to newest, capped at 16 events per scope. Parameters use AdamW with learning rate 0.001, weight decay $10^{-4}$, batch size 2,048, gradient clipping at 5, and at most 30 epochs. The checkpoint minimizes integrated Brier in an earlier calibration window, with patience 5. The same window fits the optional scalar temperature in log bounds $[-1.5, 1.5]$. Final forecasts use frozen development checkpoints and do not

benefit from additional fitting on the development outcomes.

**Dependence fitting.** At each eligible origin, the pair sampler draws 1,000 uniform-node pairs and, for each relation type, 1,000 pairs from uniformly sampled eligible groups and 1,000 pairs from groups sampled proportionally to $n_g(n_g-1)$. Groups must contain at least two eligible entities for pair fitting. This is a composite training distribution, not an unbiased all-pairs likelihood. Borg has 720,000 fitting and 240,000 validation pairs; BPI has 150,000 and 51,000. Features are standardized on fitting queries and clipped to $[-10, 10]$.

The global and static models use AdamW learning rate 0.02 and no weight decay; context models use 0.001 and weight decay $10^{-4}$. All use pair batches of 4,096, at most 50 epochs, gradient clipping at 5, and validation patience 8. Context offsets are bounded by $2\tanh(\cdot)$, with mean-squared shrinkage coefficient 0.01. Gaussian pair probabilities use 32-point quadrature in double precision. Validation cycles over the 16 horizon-pair combinations. Model selection uses validation pair likelihood; the reported final metric remains the independently specified scenario score.

**Numerical checks and hardware.** Experiments use Python 3.10.14, PyTorch 2.5.1 with CUDA 12.4, NumPy 2.2.6, SciPy 1.14.1, and two NVIDIA RTX A6000 GPUs. Local tests cover replay and label semantics, causal history retrieval, monotone trajectories, scoring, cross-chunk factor sharing, permutation controls, residual resampling, and frozen-protocol verification. A GPU sampling audit with 200,000 draws checks marginal preservation, repeated-seed determinism, and shared factors across entity chunks. These are implementation checks, not evidence that a fitted distribution is calibrated in the real system.

GPU projection aggregation uses integer counts. Independent CPU and CUDA random streams need not produce identical draws. A saved temporal-model inference audit found a maximum CPU–CUDA probability discrepancy of about 0.0021, with much smaller mean error, from floating-point time-feature arithmetic. Matching-CUDA inference reproduced saved probabilities to below $2 \times 10^{-7}$. Final temporal inference therefore uses CUDA. Runtime tables measure complete scenario sampling plus projection scoring, not isolated GPU kernel time; other workloads may share the host.

**Uncertainty calculation.** For each model pair, scores are first averaged over seeds at each origin. We resample paired chronological differences using circular moving blocks, with 10,000 replicates and bootstrap seed 20261014. Borg uses blocks of 4 and 8 origins; BPI uses 14 and 28. We report percentile 95% intervals only when at least two blocks fit within the observed span. Overlap of forecast horizons, drift, and the small number of independent periods remain limitations. No multiple-comparison-adjusted significance claim is made from exploratory tables.

**Artifact boundaries.** The source manifests contain URLs, byte counts, and SHA-256 hashes. Run manifests preserve feature definitions, data hashes, configurations, seeds, checkpoints, and scores. The final protocol freezes selected checkpoints and evaluation code before final-period preparation. Original exploratory reports are retained and are distinguishable from final reports by their period tags. Tables and figures are generated from saved results. Public raw data are obtained from the original providers under their original terms; the BPI archive is not relicensed as Creative Commons by this artifact.

The first final Borg replay stopped before producing cohort scores because its Parquet timestamps were floating-point while query timestamps were integers. An audited schema correction checks every timestamp for exact integer representation before a lossless cast, preserving the original prefix and failed logs. The frozen model and evaluation code were unchanged. This correction is recorded separately from the original freeze and is included in the reproduction instructions; it was not a model adjustment based on scores.

# B Supplementary quantitative results

Table 3: BORG final projection diagnostics. Equal horizon and origin weights; the projection CSV includes all families and modes; the released origin-score tables retain individual seeds.

| Projection/model | CRPS | Cov. | Interval score | $\text{Loss}_9$ |
|---|---|---|---|---|
| job: Independent | 0.01957 | 0.914 | 0.3073 | 0.0623 |
| job: Global | 0.01542 | 0.971 | 0.1610 | 0.0312 |
| job: Typed static | 0.01451 | 0.978 | 0.1390 | 0.0282 |
| job: Typed context | 0.01370 | 0.983 | 0.1247 | 0.0270 |
| job: Permuted incidence | 0.01729 | 0.943 | 0.2194 | 0.0439 |
| job: S Typed static | 0.01515 | 0.953 | 0.1485 | 0.0387 |
| job: Historical residual | 0.02088 | 0.909 | 0.3564 | 0.0827 |
| machine: Independent | 0.02547 | 0.965 | 0.2153 | 0.0524 |
| machine: Global | 0.02518 | 0.980 | 0.2011 | 0.0507 |
| machine: Typed static | 0.02504 | 0.976 | 0.2004 | 0.0500 |
| machine: Typed context | 0.02498 | 0.978 | 0.1995 | 0.0498 |
| machine: Permuted incidence | 0.02535 | 0.968 | 0.2105 | 0.0516 |
| machine: S Typed static | 0.02505 | 0.974 | 0.2012 | 0.0502 |
| machine: Historical residual | 0.02763 | 0.942 | 0.2571 | 0.0648 |
| population: Independent | 0.05746 | 0.100 | 1.0853 | 0.2785 |
| population: Global | 0.03877 | 0.988 | 0.2912 | 0.0833 |
| population: Typed static | 0.03807 | 0.763 | 0.2867 | 0.0802 |
| population: Typed context | 0.03821 | 0.783 | 0.2902 | 0.0856 |
| population: Permuted incidence | 0.03808 | 0.765 | 0.2873 | 0.0802 |
| population: S Typed static | 0.03919 | 0.655 | 0.3335 | 0.1028 |
| population: Historical residual | 0.04415 | 0.772 | 0.3923 | 0.1045 |

Table 4: BPI final projection diagnostics. Equal horizon and origin weights; the projection CSV includes all families and modes; the released origin-score tables retain individual seeds.

| Projection/model | CRPS | Cov. | Interval score | $\text{Loss}_9$ |
|---|---|---|---|---|
| population: Independent | 0.03421 | 0.219 | 0.5710 | 0.0674 |
| population: Global | 0.02728 | 0.869 | 0.2085 | 0.0847 |
| population: Typed static | 0.02857 | 0.668 | 0.2795 | 0.0697 |
| population: Typed context | 0.02824 | 0.720 | 0.2589 | 0.0727 |
| population: Permuted incidence | 0.02852 | 0.671 | 0.2781 | 0.0696 |
| population: S Typed static | 0.03413 | 0.322 | 0.4709 | 0.0659 |
| population: Historical residual | 0.02548 | 0.746 | 0.2476 | 0.0722 |
| handler: Independent | 0.06599 | 0.853 | 0.5660 | 0.1680 |
| handler: Global | 0.06462 | 0.911 | 0.4998 | 0.1733 |
| handler: Typed static | 0.06440 | 0.919 | 0.4920 | 0.1747 |
| handler: Typed context | 0.06436 | 0.935 | 0.4792 | 0.1789 |
| handler: Permuted incidence | 0.06536 | 0.881 | 0.5364 | 0.1692 |
| handler: S Typed static | 0.06640 | 0.892 | 0.5036 | 0.1697 |
| handler: Historical residual | 0.06225 | 0.916 | 0.5025 | 0.1794 |